6.3 Medians and Altitudes of Triangles

Name _____ Period ____ Date _____

Geometry 6.3 (30-minute plan) · Student sheet

I can…

- use the Centroid Theorem to find lengths on a median.
- find the centroid of a triangle in the coordinate plane.
- find the orthocenter of a triangle and tell where it is.

Vocabulary · fill in the missing word

Median of a triangle — a segment from a vertex to the _____ of the opposite side

Centroid — the point of _____ of the medians of a triangle

Altitude of a triangle — the _____ segment from a vertex to the opposite side or to the line that contains the opposite side

Orthocenter — the point where the lines that contain the _____ of a triangle intersect

Part 1 · Notes and examples (fill in each note, then work the example)

Centroid Theorem: the centroid is _____ of the distance from each vertex to the midpoint of the opposite side.

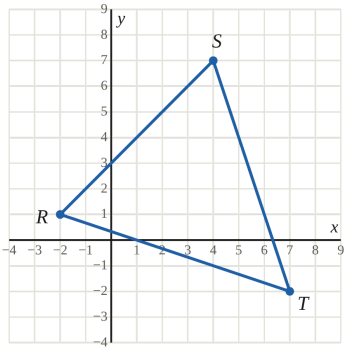
Example 1

Point P is the centroid of $\triangle LMN$, and $\overline{NQ}$ is a median. NQ = 18. Find NP and PQ.

Centroid from coordinates: the centroid's coordinates are the _____ of the vertices' coordinates: $((x_1 + x_2 + x_3)/3, (y_1 + y_2 + y_3)/3)$.

Example 2

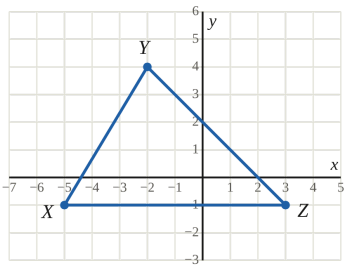
Find the centroid of the triangle with vertices R(−2, 1), S(4, 7), and T(7, −2).



Where the orthocenter is: inside an acute triangle, at the vertex of the _____ of a right triangle, and outside an obtuse triangle.

Example 3

Find the orthocenter of the triangle with vertices X(−5, −1), Y(−2, 4), and Z(3, −1). Is it inside, on, or outside the triangle?



Part 2 · Partner practice

Partner practice

Find the centroid of the triangle with vertices (1, 1), (5, 3), and (3, 8).